



Traders@UST – Quant (Basic) Team 25-26 Spring Final Project Report

A Beginner's Dive into Multi-Factor Models

Team Lead:	Oscar Tsoi	(QFIN Y2)
Team members:	Bryan Chua	(QFIN Y1)
	Jacky Liu	(GBUS Y1)
	Wilson Orlando	(DSCT Y2)

Team supervisor:	Lincoln Liang	(IS Y2)
------------------	---------------	---------

Data Sponsorship:	Massive.com
-------------------	-------------



Contents

Executive Summary	3
Introduction	4
Theory	5
Modelling returns	5
Modelling risk	6
Portfolio optimization	6
Data & Methodology	7
Factor list and evaluation	10
1. Momentum/Mean Reversion Factors	10
2. Fundamentals Factors	13
3. Liquidity Factors	14
Backtesting & Results	17
Conclusion and future plans	20
Acknowledgements	21
References	21

Executive Summary

This report details the development of a quantitative multi-factor strategy by the Traders@UST Quant Team, focused on discovering alpha signals within the US equity market.

Methodology and Strategy Design

Model: A cross-sectional linear regression framework to predict excess returns, paired with a Barra-style risk model to decompose systematic and idiosyncratic risk.

Universe: All US stocks with NaN values less than 50%

Timeframe: Past 10 years, 2016-01-01 to 2025-12-31

Alpha Signals: Focused on Momentum/Mean Reversion (short-term price corrections) and Liquidity (price impact and tradability). Fundamental factors were ultimately excluded due to data sparsity and reporting lags.

Performance highlights

Net Annual Return: 11.24% (Benchmark: 16.64%)

Sharpe Ratio: 1.246

Max Drawdown: -9.64%

Information Ratio: -0.258

Critical findings and future iterations

While the strategy remained profitable with a respectable Sharpe ratio, it underperformed the benchmark. Key bottlenecks included:

- Multicollinearity: High correlation between momentum and liquidity signals diluted predictive power.
- Execution Gaps: An overly high risk-aversion parameter led to insufficient turnover, preventing the model from fully capturing short-term alpha.
- Future Roadmap: Priority will be placed on Elastic Net regularization to handle factor selection, PCA for orthogonalization, and exploring XGBoost to capture non-linear market relationships.

Introduction

Since the introduction of stocks, people have sought many ways to predict and explain stock returns. Why do some assets perform better than others? What risks affect their returns? To answer them, early financial models, such as the Capital Asset Pricing Model (CAPM), suggested that market risk was the main factor influencing returns. However, later researchers, like Fama, French and Carhart, found that market risk alone could not fully explain differences in asset performance, leading to the development of multi-factor models. These expand the analysis by incorporating several variables, such as size, value, momentum, profitability, and investment factors, that may influence returns simultaneously. While these models were originally used to explain asset pricing and manage portfolio risk, they are now also widely applied in quantitative investing as a framework for discovering alpha signals.

In practice, many quantitative funds, proprietary trading firms, and brokerage research teams use multi-factor models to search for predictive factors that may generate excess returns. They test a wide range of signals derived from price, volume, volatility, financial statements, analyst expectations, alternative data, and market microstructure. Firms such as WorldQuant and many Chinese securities brokerages have popularized large-scale factor mining, where researchers generate and evaluate thousands of candidate signals using statistical and machine learning techniques. Promising factors are then combined into diversified alpha models to improve forecasting accuracy and reduce dependence on any single signal.

Our final project thus focused on the discovery of these alpha signals, and building a linear regression model to combine these factors. Using the WorldQuant Brain platform, we generated and tested a range of candidate signals based on market data and factor expressions. The platform allowed us to evaluate each signal using performance measures such as return, Sharpe ratio, turnover, and robustness across different market environments. Through this screening process, we were able to identify several factors with relatively strong and stable predictive power, which were then fed into our own evaluation code for further analysis. We then applied a linear regression model to estimate the returns and risk of each stock and used this to optimize a portfolio on the entire US stock universe.

This project does not aim to find a good strategy. In fact, our strategy is far from being robust and executable, and there are many ways we can improve our model given more time. Rather, as beginners with limited time, we approached this project as a valuable learning opportunity to apply our knowledge of statistics, data analysis, and financial markets in a practical setting. By exploring the process of factor discovery, model construction, and performance evaluation, we hope to take our first steps into the field of quantitative research and trading while gaining a deeper understanding of how systematic investment strategies are developed.

Theory

Modelling returns

The ultimate goal is to predict excess stock returns R_{it} for security i at time t over the risk-free rate, using *factor exposures* X_{it} (abbreviated as *factors*) which are known at time t .

Suppose we have n assets and k factors. We use a linear regression model below to predict:

$$R_{it} = \sum_{j=1}^k f_j X_{(i,j)t} + u_{it}$$

Where f_j is the *factor return* (i.e., the regression coefficient) of factor j , $X_{(i,j)t}$ is the *factor exposure* of factor j of security i at time t , and u_i is the *specific return* (i.e., the alpha) of security i .

u_i essentially captures the part of returns that cannot be explained by our k factors.

By dropping time in our equation and converting it to matrix form, this translates into:

$$\vec{R} = X\vec{f} + \vec{u}$$
$$\begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_n \end{pmatrix} = \begin{pmatrix} X_{1,1} & X_{1,2} & \cdots & X_{1,k} \\ X_{2,1} & X_{2,2} & \cdots & X_{2,k} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n,1} & X_{n,2} & \cdots & X_{n,k} \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{pmatrix} + \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}$$

This forms a multi-linear regression model resembling the form $\vec{Y} = X\vec{\beta}$ with the constant term being $\vec{u}$.

Therefore, we have that $E(\vec{R}) = XE(\vec{f}) + E(\vec{u})$, and we find $E(\vec{f})$ by regressing over every timestamp and taking the average of their factor returns.

For the CAPM model (Sharpe, 1964), there is only 1 factor: the market risk premium $R_M - R_F$. Fama and French (1993) expanded it to include 2 more: *Small minus Big Size* SmB , and *High minus Low M/B ratio* HmL . Carhart then added the momentum factor *Up minus Down* UmD , and MSCI Barra expanded it into 12 *style* factors in its US Equity Model (USE4) (Menchero et al., 2011), like beta, size, volatility, book-price ratio, earnings yield, etc.

Our multi-factor model thus imitates Barra's model but for the purpose of alpha generation. Instead of using the style factors, we aim to predict stock returns better than Barra under a shorter timeframe and capture alphas that are not yet realized by the market.

Modelling risk

Using our model for estimating returns, we can model risk by finding the variance of the returns.

$$\text{Var}(R_i) = \text{Var}(\bar{X}_i \cdot \vec{f} + u_i) = \text{Var}(\bar{X}_i \cdot \vec{f}) + \text{Var}(u_i)$$

By making it in matrix form and doing some linear algebra, we have:

$$\Sigma = XF\bar{X}^T + \Delta$$

Where $F = \begin{pmatrix} \sigma_{f_1}^2 & \sigma_{f_1, f_2} & \cdots & \sigma_{f_1, f_k} \\ \sigma_{f_2, f_1} & \sigma_{f_2}^2 & \cdots & \sigma_{f_2, f_k} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{f_k, f_1} & \sigma_{f_k, f_2} & \cdots & \sigma_{f_k}^2 \end{pmatrix}$ which is the *factor return covariance matrix*,

and $\Delta = \begin{pmatrix} \sigma_{u_1}^2 & 0 & \cdots & 0 \\ 0 & \sigma_{u_2}^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{u_k}^2 \end{pmatrix}$ which is a diagonal matrix containing the specific, or

idiosyncratic, risk variances of each stock.

This decomposition separates total risk into two components: systematic risk explained by common factors, and stock-specific risk that is not captured by those factors. In our implementation, the factor covariance matrix F is estimated from historical factor returns using an exponentially weighted scheme so that more recent observations receive greater weight in the risk forecast. The final predicted covariance matrix Σ can then be used.

In practice, we adopt a Barra-style risk model. Our factor exposure matrix includes the style factors used elsewhere in the model as well as 49 industry indicator variables, which are implemented using SIC-code-based industry classification as an approximation to Barra industry factors. These industry variables are not used in our expected return prediction because the alpha factors were already neutralized at the subindustry level. However, they remain important in the risk model because stocks in the same industry tend to exhibit correlated residual movements, and excluding industry structure would underrate covariance risk.

Portfolio optimization

Using the calculated expected returns and risk, we can optimize a portfolio by maximizing the utility function:

$$U = E(R) - \frac{1}{2}A\sigma^2$$

With A being the risk aversion. Here, we take 2.5 as the risk aversion parameter for a proxy of the market risk aversion.

Data & Methodology

Data sources

We obtained our data from Massive.com, who generously sponsored us with their Stocks Advanced plan for use throughout the entire project.

The following are our sources:

- OHLCV & VWAP: `/v2/aggs/grouped/locale/us/market/stocks/{date}`
- Fundamentals:
 - Balance sheet: `/stocks/financials/v1/balance-sheets`
 - Income statement: `/stocks/financials/v1/income-statements`
 - Cash flow statement: `/stocks/financials/v1/cash-flow-statements`
 - Ticker details (SIC code, CIK): `/v3/reference/tickers/{ticker}`

Data cleaning

1. Remove holidays where OHLCV data is not available
2. Remove non-stocks like ETFs through filtering such that ticker type is common stock.
3. Remove stocks with too many NaN values:
If the number of NaN values of its OHLCV data exceeds 50% of its total data, then omit.
4. Forward imputation of fundamentals data:
 - Since the obtained fundamental data is presented in the form of SEC filings, when inputting into the main dataframe, forward imputation is used to assume that in between reporting dates, the fundamentals follow the last reported data.
 - As for OHLCV data, since some companies might have delisted and thus do not have such data today, they should not be forward imputed.
5. Backward imputation of SIC code and CIK. We assume that SIC codes and CIK do not change with time.
6. Standardizing fundamental data according to timeframe, as some fundamental data is annual and some are quarterly. Here we standardize to quarterly by dividing income statement data (i.e., 'operating_income', 'ebitda', etc.) by 4.

Factor processing

After obtaining raw factor values from the raw factor formulas, we perform the following processing steps:

1. Winsorization: Replacing outliers outside of lower and upper quantile bounds with the boundary values
2. Neutralization: Centering all factor values around group means
Here we have the option to centralize to market mean, industry mean, and sub-industry mean
3. Decay: Replacing each factor value with the weighted average of the previous n days factor values, using linear weights and weighing more on current days. The formula is as follows:

$$X_t = \frac{nX_t + (n-1)X_{t-1} + (n-2)X_{t-2} + \dots + X_{t-n+1}}{1 + 2 + \dots + n}$$

4. Cross-sectional normalization: Replace all factors with their cross-sectional zscores for scaling.

Factor evaluation

We first test them on WorldQuant Brain, which turns our factors into weights in its all-stock portfolio. Stocks with higher factor values are overweighted in long positions, and stocks with lower factor values are overweighted in short positions. Promising factors with good Sharpe ratio and relatively low turnover ratio are selected as factors in our multi-factor model.

We then evaluate the factors based on their mean information coefficient (IC) and IC information ratio (ICIR, i.e., standard deviation of the IC). The formulas are as follows:

$$IC_t = \text{Corr}(\hat{R}_t, R_t)$$
$$ICIR = \frac{\text{Mean}(IC_t)}{\text{Stdev}(IC_t)}$$

IC_t is essentially the Pearson correlation coefficient between the predicted returns $\hat{R}_t$ and actual returns R_t at time t .

A high IC thus typically means that the factor predicts stock returns well, and a high ICIR typically means that the factor predicts well and is stable across time.

Regression method

Time-series vs cross-sectional regression

Traditionally, $\vec{f}$ was predicted using time-series regression, i.e., regressing each stock over the entire historical period to obtain the betas of each stock. However, doing so undermines the short-term nature of some short-term factors (such as mean reversion). This is why we use cross-sectional regression instead and take the factor and residual returns as the average of all the predicted factor and residual returns, so that the weighting can immediately change after events captured by our alphas.

(Regularization and feature selection were not finished due to time constraints)

L1 regularization, L2 regularization vs Elastic net regularization

We decided to use elastic net regularization to select pairs of factors to combine and use L2 regularization in our final model to reduce overfitting. However, since our model only has a handful of factors, L2 regularization might not be necessary.

Our method for tuning the regularization hyperparameters rely on a walk-forward cross validation process using the “sliding window” method, where the model is trained on a starting block of historical data, cross-validated on the immediate next time period, and the training window is shifted forward and the process is repeated. We repeat this whole process on different potential hyperparameter values and calculate the average MSE for each potential value, selecting the one with the lowest MSE. The grid we decided to use was a logarithmic grid of values [0.001, 0.01, 0.1, 1.0, 10.0], and for elastic net, the outer product of this list.

Orthogonalization

Our goal with our model was to ensure that it can still be explainable, i.e., we can explain the factors’ contributions to the model and what effects a certain event might have on the model for risk management. Therefore, we felt the need to avoid orthogonalizing all factors as this would obfuscate the factors’ nature. However, if there are factors that are highly correlated with one another, we may use principal component analysis (PCA) to orthogonalize these factors, ensuring that the resulting “de-correlated” factor may still be explainable.

Factor list and evaluation

We have found numerous alphas to construct our strategy. In this part, we are going to elaborate on the rationale behind these alphas. The performance of these factors are listed in page

1. Momentum/Mean Reversion Factors

0. (Not tested on WorldQuant Brain)

$$-rank(ts_zscore(returns, 252))$$

Logic: Find companies with an abnormal price surge/plunge within 1 year and short/long it.

1.

$$-zscore((close - vwap)/ts_mean(vwap, 20)) * rank(ts_delta(volume, 3)) * rank(cap)$$

Logic: Find large-cap companies' stock with intraday price and volume surge as they tend to go in the reverse direction.

Signal 1: Closing Price Deviation from Intraday Fair Value

- Compares the closing price to the volume-weighted average price (VWAP) of the day, then scales down this difference by the 20-day average of VWAP to make it comparable across stocks.
- Generate overweight signal when the close is *below* VWAP.
- The resulting ratio is then standardized across stocks (z-score), measuring how many standard deviations the risk-adjusted discount to VWAP is from the cross-sectional mean.

Signal 2: Short-Term Volume Expansion

- Measures recent change in trading volume over a very short window and ranks it across all stocks.
- A high rank indicates the stock has experienced an unusually large surge in volume compared to others.

Signal 3: Market Capitalization

- Ranks stocks by their market capitalization.
- A higher rank means a larger, more established company.

2.

$$\text{rank}(\text{ts_zscore}(-\text{close} / \text{ts_mean}(\text{close}, 20), 5)) + \text{rank}(\text{ts_rank}(\text{eps} / \text{close}, 60))$$

Component 1: Short-term Oversold Bounce Logic

- The signal strengthens when the price falls below its 20-day moving average, reflecting the degree of short-term undervaluation relative to the trend.
- It standardizes this deviation over the past 5 days to gauge whether the current oversold condition is unusually extreme compared to very recent history.
- A cross-sectional comparison then favours stocks where the selling pressure has been most abnormal and the decline most aggressive in the short run.
- Core idea: after a panic-driven sharp drop, prices tend to stage a short-term mean-reverting bounce.

Component 2: Medium-term Relative Valuation Logic

- Earnings yield (earnings per share divided by price) serves as a direct valuation gauge — a higher yield means a cheaper valuation.
- By ranking the current earnings yield against its own 60-day history, a higher rank indicates the stock is trading near its cheapest level over that period.
- A subsequent cross-sectional ranking then overweights stocks that are cheap relative to their own recent valuation range.
- Core idea: valuations exhibit mean-reversion, and a persistently rising earnings yield may signal improving profitability not yet fully priced in by the market.

3.

$$\text{rank}(\text{ts_mean}(\log(\text{open} / \text{ts_delay}(\text{close}, 1)), 20)) - \text{rank}(\text{ts_mean}(\log(\text{close} / \text{open}), 20))$$

Formula breakdown:

- Subtracts the cross-sectional rank of the 20-day average intraday return from the 20-day average overnight return.
- The result is a long-short score: high values favour stocks with strong persistent overnight gains and weak persistent intraday gains, while low values indicate the opposite.

Long Leg: Strong Overnight, Weak Intraday

- Positive overnight returns (close-to-open) are often driven by institutional and fundamental positioning, which tends to exhibit persistence.
- Negative intraday returns (open-to-close) suggest that during market hours the stock is being sold into temporary noise or sentiment, creating relative value.
- Together, a consistently high overnight return and low intraday return point to an underlying bid that is masked by day-session noise — a setup for continued upward drift.

Short Leg: Weak Overnight, Strong Intraday

- Stocks with weak persistent overnight returns may lack fundamental sponsorship.
- Yet if intraday returns are persistently strong, this often reflects short-term speculative chasing that is likely to fade.
- Ranking low on overnight strength and high on intraday strength identifies names vulnerable to mean-reversion on the next open.

Core Economic Rationale

- The strategy exploits a separation in return drivers: overnight moves reflect informed, longer-horizon demand; intraday moves are more susceptible to sentiment and mean-reverting flows.
- By isolating stocks where these two components diverge over a 20-day period, the signal captures a premium for providing liquidity to overnight positioning while avoiding intraday momentum that tends to reverse.

4.

$$N = 60; \text{amp} = (\text{high} / \text{low}) - 1; \text{price_change} = \text{close} / \text{ts_delay}(\text{close}, 1) - 1; - \\ \text{ts_sum}((\text{ts_zscore}(\text{amp}, N) > 1? \text{price_change} : 0), N)$$

High-volatility day detection: Uses the intraday amplitude ($\text{high}/\text{low} - 1$) and flags days where its z-score over a 60-day window exceeds 1 — i.e., days when the day's range is unusually large relative to its own recent history. On those flagged days, records the daily return. Summing up those daily returns over the past 60 days, then takes the negative of that sum.

Interpretation:

- **Positive alpha value:** Over the past 60 days, price mostly fell on high-volatility days, suggesting overselling and a potential upward reversal.
- **Negative alpha value:** Over the past 60 days, price mostly rose on high-volatility days, suggesting overbuying and a potential downward reversal.

Behavioral rationale: Abnormally wide ranges often reflect emotional overreaction. The subsequent reversion of those moves implies that such days mark temporary price extremes. The 60-day window captures a medium-term reversal effect, aggregating signals from multiple overreaction episodes to form a stable contrarian score.

5.

$$-1 * ((rank(ts_delta(close, 5) / ts_mean(high - low, 5)) - 0.5) ** 3)$$

Rationale:

- Measures risk-adjusted 5-day return. Converts to a cross-sectional rank and centers it around zero.
- Cubes the centered rank to heavily amplify only extreme overbought or oversold readings, suppressing mid-range signals.
- Applies a negative sign to bet against extremes: long deeply oversold stocks, short strongly overbought ones.
- Profits from the tendency of sharp, volatility-adjusted swings to reflect overreaction and subsequently reverse.

2. Fundamentals Factors

1.

$$-zscore(ts_mean((enterprise_value / ebitda), 252) / (enterprise_value / ebitda))$$

Rationale:

- Compares current EV/EBITDA to its own 1-year rolling average.
- A higher ratio, i.e., current cheaper than history, becomes a positive signal after z-scoring and negation.
- The negative z-score ensures the alpha favours stocks undervalued relative to their own past, and shorts those overvalued.
- Captures valuation mean-reversion: multiples tend to revert toward their long-term norms.

2.

$$oey = operating_income / cap; num_days_neg = ts_sum((returns < 0 ? 1 : 0), 252);$$

$$rank(ts_delta(oey, 125) / ts_std_dev(oey, 252)) + rank(num_days_neg) + rank(-ts_delta(close, 1))$$

Rationale:

- Combines three equal-weighted signals: improving profitability trend, persistent pessimism, and a one-day reversal.
- **First signal:** measures the risk-adjusted trend of operating income yield over the past half-year. A rising yield suggests fundamentals are strengthening in a sustained way.
- **Second signal:** counts the number of down days over the past year. More down days reflect chronic underperformance and potential over-pessimism, creating a contrarian long opportunity.

- **Third signal:** rewards a recent daily price drop by ranking the negative of the last day's return. Captures a short-term oversold bounce effect.

3. Liquidity Factors

1.

$$- \text{abs}(\text{returns}) / (\text{volume} * \text{close})$$

Rationale:

- Measures the price impact per unit of dollar volume.
- A high value indicates that a given price move occurred on relatively thin liquidity, suggesting potential for price instability or overreaction.
- The alpha favours stocks with low impact and penalizes those with high impact.
- Captures the idea that price changes in illiquid stocks tend to be less reliable and more prone to reversal or manipulation.

2.

$$\text{zero_return_days} = \text{ts_sum}(\text{returns} == 0 ? 1 : 0, 252); -\text{zero_return_days}$$

Rationale:

- Counts the number of zero-return days over the past year.
- A high zero-return count signals illiquidity, lack of investor interest, or stale pricing.
- The alpha penalises stocks with many flat days and favours those with continuous price discovery.
- Serves as a liquidity and tradability filter: it avoids names where price signals are unreliable or expensive to trade.

3.

$$\text{illiquid} = \text{rank}(\text{ts_sum}(\text{returns} == 0 ? 1 : 0, 60)); \text{rev_neutral} = \text{group_rank}(\text{ts_mean}(\text{vwap}, 10) - \text{close}, \text{subindustry}) - 0.5;$$

$$\text{healthy_trend} = \text{rank}(\text{close} / \text{ts_mean}(\text{close}, 60)); \text{illiquid} * \text{rev_neutral} * \text{healthy_trend}$$

Rationale:

- **Illiquidity signal:** Counts zero-return days over 60 days to identify low-activity, neglected stocks where mispricing is more likely and reversal potential is larger.
- **Industry-relative mean reversion:** Within each sub-industry, ranks how far VWAP over 10 days sits above the current close. A positive rank means the stock is trading at a discount to its recent VWAP, signalling a short-term bounce opportunity.
- **Trend filter:** Ranks the ratio of close to its 60-day average, ensuring the stock is in a medium-term uptrend before taking a long reversal bet.

- **Multiplication logic:** All three components must align — illiquid, oversold within its sector, and in a healthy uptrend — so the alpha concentrates on under-followed names that are temporarily cheap but moving upward.

Evaluation metrics

(MR = Mean reversion, F = Fundamental (V=value, Q=quality), L = Liquidity, B = Barra Factor)

WorldQuant Brain metrics

	Sharpe ratio	Turnover ratio	Return	Drawdown
MR 0*	-	-	-	-
MR 1	1.71	59.44%	25.00%	11.16%
MR 2	2.66	38.20%	10.69%	2.58%
MR 3	1.10	16.86%	6.12%	3.69%
MR 4	1.40	12.48%	18.48%	26.29%
MR 5	1.72	71.63%	16.38%	7.16%
MR B*	-	-	-	-
F 1	1.22	19.73%	31.59%	26.22%
F 2	1.26	13.22%	8.82%	7.72%
V B*				
Q B*				
L 1	1.16	33.85%	15.26%	10.95%
L 2	1.18	3.06%	10.67%	17.84%
L 3	1.77	33.29%	12.18%	4.88%
L B*				

* not tested on WorldQuant Brain

IC, ICIR

	1 day		5 days		10 days		30 days	
	IC	ICIR	IC	ICIR	IC	ICIR	IC	ICIR
MR 0	0.009252	0.157487	0.014736	0.264651	0.015687	0.285348	0.019225	0.427537
MR 1^	0.00561	0.025658	0.001435	0.006847	0.000085	0.000428	-0.03448	-0.19543
MR 2^	-0.03009	-0.13256	-0.05228	-0.23187	-0.06314	-0.27882	-0.11821	-0.51180
MR 3	0.003197	0.069254	0.005336	0.108691	0.006777	0.145744	0.011263	0.233473
MR 4	0.007274	0.110497	0.009745	0.139572	0.011969	0.16254	0.012069	0.227237
MR 5	0.017788	0.290478	0.016057	0.283658	0.010341	0.222123	0.004929	0.127802
MR B	0.02027	0.234903	0.020884	0.244335	0.013824	0.171939	0.005513	0.085909
F 1^	0.00157	0.003322	-0.01452	-0.03093	-0.01494	-0.03258	-0.06855	-0.14748
F 2^	-0.05034	-0.19331	-0.08464	-0.30047	-0.09242	-0.34197	-0.16307	-0.65273
V B^	-0.00148	-0.01830	-0.00225	-0.02464	0.000075	0.00083	0.014114	0.161439
Q B^	0.004524	0.042871	0.014424	0.118649	0.021418	0.183544	0.032764	0.309774
L 1	-0.00368	-0.05553	-0.00287	-0.05054	0.000289	0.005877	-0.00176	-0.03705
L 2	-0.00104	-0.01829	-0.00864	-0.15526	-0.01682	-0.27517	-0.04149	-0.58772
L 3	0.008382	0.156183	0.01379	0.271553	0.014118	0.326746	0.01255	0.316218
L B	0.001895	0.030569	0.005718	0.082345	0.00587	0.078402	0.000911	0.018374

^ Unfortunately, the fundamental data obtained was sparse and could not act as a reliable source of information for these factors. Therefore, these factors were not included in the final model for the time being. Further work needs to be done to obtain and clean these fundamental data.

Notable factors

MR 0: Comparatively and consistently high ICIR for all timeframes

MR 5: Great ICIR on shorter timeframes (1d, 5d)

L 2: Comparatively high ICIR on 5d dataframe, great ICIR on longer timeframes (10d, 30d)

L 3: Great ICIR across all timeframes

Backtesting & Results

Final strategy

We decided to exclude factors using fundamental data because of two reasons:

1. Some factors (e.g., fundamental_2) produce high variation, possibly due to lack of fundamental data for a majority of stocks.
2. Fundamental factors typically work well over longer periods as their reporting period often spans months or years. This will require longer holding periods and including them in a short-term model might not be feasible.

For selecting the holding period, we used 5 days because the factors (especially the notable ones) typically have higher ICIR on 5-day timeframe, and because the rationale of our factors mainly run on shorter timeframes. Furthermore, a 5-day timeframe balances well between sample size and predictability of factors.

When performing cross-sectional regression on returns, we have also found that the average adjusted R-squared for 5 days was the highest, while RMSE was the 2nd lowest (after 1 day RMSE).

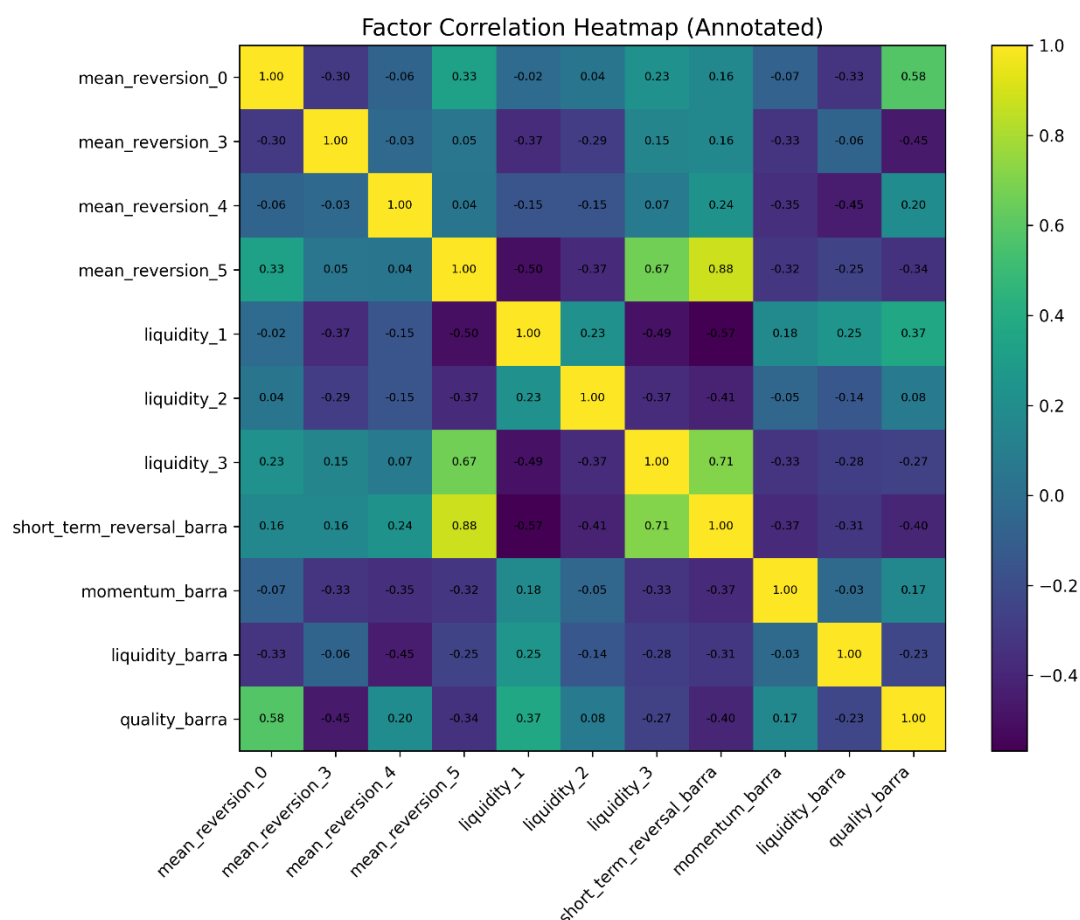
Average adjusted R² across holding periods

Holding Period	Mean adj R ²	Std adj R ²	Count
1 day	0.040250	0.038006	1,235
5 days	0.042697	0.044222	1,239
10 days	0.040385	0.040601	1,234
30 days	0.038707	0.034449	1,214

Average RMSE across holding periods

Holding Period	Mean RMSE	Std RMSE	Count
1 day	0.047816	0.288686	1,243
5 days	0.168505	0.695811	1,239
10 days	0.301101	0.977425	1,234
30 days	0.737620	1.647154	1,214

Correlation matrix of factors (checking multi-collinearity)

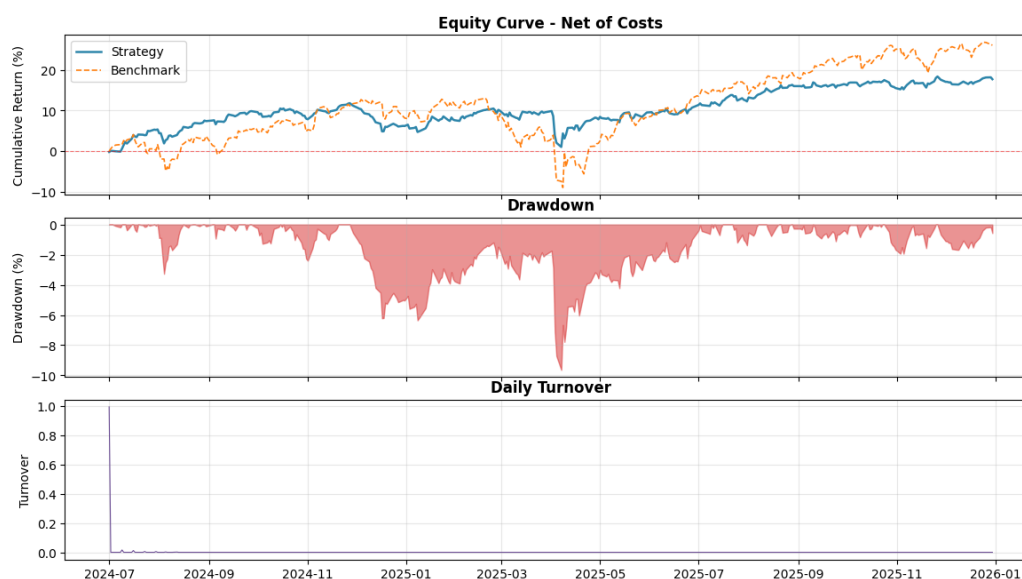


Notable findings:

- MR B (*short_term_reversal_barra*), L 1, L 3 and MR 5 show strong correlation with one another. These 4 need to de-correlate through orthogonalization (i.e., principal component analysis)
- Quality_barra and MR 0 show relatively strong correlation

Backtesting results on test set

Here we use the Russell 3000 index as our benchmark as our portfolio runs on around the top 3000 stocks in the US.



Total return	17.69% (vs benchmark 26.05%)
Annual return (net)	11.24% (vs benchmark 16.64%)
Annual return (gross)	11.41%
Cost drag (annual)	0.17%
Annual volatility	9.02%
Sharpe ratio	1.246
Sortino ratio	1.511
Max drawdown	-9.64%
Ulcer index	0.0235
UPI	4.786
Average daily turnover	0.003
Periods	379

Vs Benchmark

Active annual return	-5.05%
Tracking error (annual)	19.55%
Information ratio	-0.258

Unfortunately, this means we were not able to beat the benchmark and our strategy needs further refinement.

Conclusion and future plans

We believe our strategy performed worse than the market because of the following reasons:

1. Lack of regularization and factor selection:

The correlation matrix shows that some factors are highly correlated, yet were included together in the model, leading to multi-collinearity issues. Lack of regularization also would suggest that the model might have overfitted, but further backtests need to be done on the training and cross-validation set to prove this is the case.

2. Setting the risk aversion too high:

As shown from the turnover graph, the turnover stayed abysmally low, which meant that the portfolio was not being actively re-weighted every 5 days. This could mean that the model was taking riskless bets which resulted in reliance on the market portfolio.

3. Lack of reliable fundamental data which resulted in some factors not being able to be included in the model

4. Factors had weak predictive signal:

All our factors were not able to achieve an IC of over 0.05 and an ICIR of over 0.5, which meant that our overall model fundamentally suffered from lack of predictive power.

5. Lazy NaN handling:

We set a NaN handling threshold of 0.5 to eliminate stocks without much data, but we might have accidentally excluded those that went bankrupt or delisted at the dataset start date, and those that were just listed not long before the dataset end date. We originally planned to have an actual filtering of the top 3000 stocks (based on market cap) but was scraped due to computation limitations. Potential solutions for this might be to either obtain Russell 3000 index tickers, or optimize our code to run the stock ranking quicker with smarter settings.

6. Model limitations:

Currently, we are using linear regression which might not capture non-linear relationships for certain factors. More advanced models such as XGBoost might work better on our factors.

Further tests to perform to evaluate where the strategy failed include:

1. Inspecting industry and sub-industry concentration to see whether the strategy biased towards certain industries, leading to concentration risk.
2. Performing the entire factor-selection and backtest using sub-universes of the US top 3000, such as the US top 1000, US top 500, etc., to check whether the strategy has survivorship bias.

Therefore, our future to-do list for this project would be:

1. Test more alpha and risk factors, and improve upon existing ones
2. Perform regularization and PCA on highly correlated clusters
3. Clean fundamental data using the aforementioned methods, and either obtain more data for stocks without fundamental data, or explore imputation methods to fill in NaN values
4. Explore other models to possibly replace linear regression models.

Acknowledgements

- Massive.com: For providing data sponsorship for use throughout this project
- Our supervisor Lincoln Liang: For his immense knowledge support and valuable advice
- Jerry Wong: For coordinating the massive.com sponsorship and providing advice on data usage

References

Sharpe, W. F. (1964). Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk. *Journal of Finance*, 19(3), 425–442.

Menchero, J., Orr, D. J., & Wang, J. (2011). *The Barra US Equity Model (USE4): Methodology notes*. MSCI Barra.

MSCI Barra. (2007). *Barra risk model handbook*.